Chapter 3 : Series

We saw series in the initial lectures and saw they were useful for integrating difficult functions. Further, infinite series are used to calculate special irrational numbers, such as π. These series are often not at all clear. For example,

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)!\,(1103 + 26390k)}{(k!)^4\, 396^{4k}}.$$

Defn 2.1 : Series

Let (a_k) be a sequence. An <u>infinite series</u>, or just <u>series</u>, is a formal expression of the form

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \cdots .$$

The <u>sequence of partial sums</u> (S_n) of a series is then defined by

$$S_n = a_1 + a_2 + \cdots + a_n$$

for each $n \in \mathbb{N}$.

Defn 3.1 : Convergent/Divergent Series

A series $\sum_{k=1}^{\infty} a_k$ is said to be <u>convergent</u> if the sequence of partial sums (S_n) converges. If (S_n) converges to $L \in \mathbb{R}$, the series is said to converge to L and is written as

$$\sum_{k=1}^{\infty} a_k = L .$$

The number $L \in \mathbb{R}$ is called the <u>sum</u> of the series. If (S_n) diverges, then we say the series diverges.

<u>Ex</u>: The series given by $\sum_{k=1}^{\infty} 1/2^k$ converges to 1.

<u>Proof</u>: The sequence of partial sums (S_n) is given by

$$S_n = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} .$$

It follows by proof by induction that

$$S_n = \frac{2^n - 1}{2^n} \Rightarrow S_n = \frac{1 - 1/2^n}{1}.$$

Since $\lim\left(\frac{1}{2^n}\right) = 0$, it follows by the algebra of limits that $\lim(S_n) = 1$. ☐

Theorem 3.3 : Geometric Series

A series is called geometric if it is of the form

$$\sum_{k=0}^{\infty} ar^k$$

with $r, a \in \mathbb{R} \setminus \{0\}$. Then a geometric series is convergent iff $|r| < 1$.

Proof: The sequence of partial sums (S_n) is given by

$$S_n = a + ar + ar^2 + \ldots + ar^n.$$

If $r = 1$, the series is clearly divergent. Otherwise, if

$r \neq 1$, since

$$(1-r)(a + ar + \cdots + ar^n) = a - ar^{n+1}$$

it follows that

$$S_n = \frac{a - ar^{n+1}}{1 - r}.$$

If $|r| < 1$, it follows that $\lim (r^{n+1}) = 0$.

Hence, by the algebraic limit theorem,

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}.$$

Otherwise, if $|r| > 1$, the series diverges as the sequence (c_n) with $c_n = r^{n+1}$ diverges. ☐

Ex: Convergent Series

The series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent.

Proof: The sequence of partial sums (S_n) is given by

$$S_n = 1 + \frac{1}{4} + \dots + \frac{1}{n^2}.$$

Since, $0 \leq 1/k^2$ for all $k \in \mathbb{N}$, it's clear that (S_n) is monotone increasing. Now,

$$S_n = 1 + \frac{1}{2 \cdot 2} + \frac{1}{3 \cdot 3} + \dots + \frac{1}{n \cdot n}$$

$$< 1 + \frac{1}{2 \cdot 1} + \frac{1}{3 \cdot 2} + \dots + \frac{1}{n \cdot (n-1)}$$

$$= 1 + \left(1 - \tfrac{1}{2}\right) + \left(\frac{1}{2} - \tfrac{1}{3}\right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$$= 1 + 1 - \frac{1}{n}$$

$$< 2.$$

Since (S_n) is monotone increasing and bounded above by 2, by the monotone convergence theorem, (S_n) converges. ☐

In this case (S_n) does not converge to 2, in fact it converges to $\pi^2/6$.

Theorem 3.5 : (Algebraic limit Theorem for Series)

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergent series with

$\sum_{k=1}^{\infty} a_k = \alpha$ and $\sum_{k=1}^{\infty} b_k = \beta$. Then,

i) $\sum_{k=1}^{\infty} (a_k + b_k) = \alpha + \beta$ $\left(\sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} (a_k + b_k) \right)$

ii) $\sum_{k=1}^{\infty} c a_k = c\alpha \quad \forall c \in \mathbb{R}$.

Proof : Exercise .

4- Series Tests

Here, we want to build up a collection of tests we can perform on (a_k), coming from a given series $\sum_{k=1}^{\infty} a_k$, that tells us something about whether $\sum_{k=1}^{\infty} a_k$ converges / diverges.

Theorem 4.1: Cauchy Criterion for Series

A series $\sum_{k=1}^{\infty} a_k$ is convergent iff given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for all $n > m > N$.

Proof: The difference of the sequence of partial sums, with $n > m$, is given by

$$S_n - S_m = a_{m+1} + a_{m+2} + \cdots + a_n$$

($\Rightarrow$) Let (S_n) be convergent. Hence, (S_n) is Cauchy.

Then, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$|S_n - S_m| = |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for all $n > m > N$.

($\Leftarrow$) Suppose if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$ for all $n > m > N$, then (s_n) is Cauchy. Hence, (s_n) is convergent. ☐

Theorem 4.2 : Divergence Test

If the series $\sum_{k=1}^{\infty} a_k$ converges then $\lim(a_k) = 0$.

Proof : Letting $n = m+1$. Then by the Cauchy Criterion for series, it follows that for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_{m+1}| < \varepsilon$$

for every $m > N$. Hence, $\lim(a_n) = 0$. ☐

Ex : i) $\sum_{k=1}^{\infty} (-1)^k$. Let $a_k = (-1)^k$. Clearly $\lim(a_k)$ doesn't exist. Hence, by the divergence test the series diverges. (Grandi's Series)

iii) $\sum_{k=1}^{\infty} \frac{k}{k^2+1}$. Here $a_k = \frac{k}{k^2+1}$ and $\lim(a_k) = 0$.

Here, the divergence test tells us nothing.

Ex: (Harmonic Series)

The series given by

$$\sum_{k=1}^{\infty} \frac{1}{k}$$

is called the Harmonic series. The harmonic series is divergent.

Proof: The sequence of partial sums (S_{2^n}) is given by

$$S_{2^n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)$$

$$+ \cdots + \left(\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}\right)$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots$$

$$+ \left(\frac{1}{2^n} + \cdots + \frac{1}{2^n}\right)$$

$$= 1 + \frac{1}{2} + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) + \dots + \left(\frac{1}{2}\right)$$

$$= 1 + \frac{n}{2}.$$

Since, (S_{2^n}) is unbounded it follows it is divergent. ∎

Theorem 4.5: (Cauchy Condensation Test)

Let (a_k) be monotone decreasing sequence with $a_k \geq 0$ for all $k \in \mathbb{N}$. Then the series $\sum_{k=1}^{\infty} a_k$ converges iff $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Outline Proof: (Not examinable)

Construct sequence of partial sums for $\sum_{k=1}^{\infty} a_k$ and for $\sum_{k=0}^{\infty} 2^k a_{2^k}$. See notes of full proof.